Predictive Modelling

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1: Key features of economic time series

Outline

Key concepts in this lecture:

- What is a time series?
- What is time series analysis?
- Why is time series analysis important?
- How do time series data typically look like?
- The modelling cycle

What is a time series?

A time series consists of observations for a certain variable y, which are measured at different points in time. The observation made at time t is denoted as y_t.

If you are a marketing manager, you may want to consider weekly sales of a product (to measure the effect of a certain promotion or price discount).

If you are an investor, you may want to examine the monthly returns of your asset portfolio.

If you are a government official, you may want to check the unemployment rate every quarter (to examine the effect of a new policy).

If you are an environmentalist, you may want to consider the (average) annual temperature (to see whether global warming is really happening).

What is a time series?

A time series consists of observations for a certain variable y, which are measured at different and sequential points in time.

The observations are ‘equally spaced’, that is, the time between consecutive observations is always the same.

A time series consists of observations y_t with $t = 1,2,3, \ldots, T$ (‘now’). T is the sample

size, or the length of the time series.

Time series data are observed at a certain sampling frequency. Sampling frequencies vary widely, from minute-by-minute, daily, monthly, quarterly, up to annually, or more.

What is time series analysis?

For many economic time series variables, consecutive observations often are closely related.

Observations from the recent past are informative for the value of current (and future!) observations.

Time series analysis involves the construction of a model, which adequately summarizes the dependence between consecutive observations.

Time series models take the general form

$$y_t = g(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta) + \varepsilon_t$$

where $g(.)$ is a function, θ denotes a vector of unknown parameters, and ε_t is an unobserved random variable, called shocks or news.

The main idea of forecasting

Take again:

$$y_t = g\left(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta\right) + \varepsilon_t$$

One-step-ahead realization from T onwards is

$$y_{T+1} = g\left(y_T, y_{T-1}, \dots, y_{T-(p-1)}; \theta\right) + \varepsilon_{T+1}$$

The forecast is based on

$$y_{T+1|T} = g\left(y_{T|T}, y_{T-1|T}, \dots, y_{T-(p-1)|T}; \theta\right) + \varepsilon_{T+1|T}$$

At time T the news at $T+1$ is unknown, so set

$$\varepsilon_{T+1|T} = 0$$

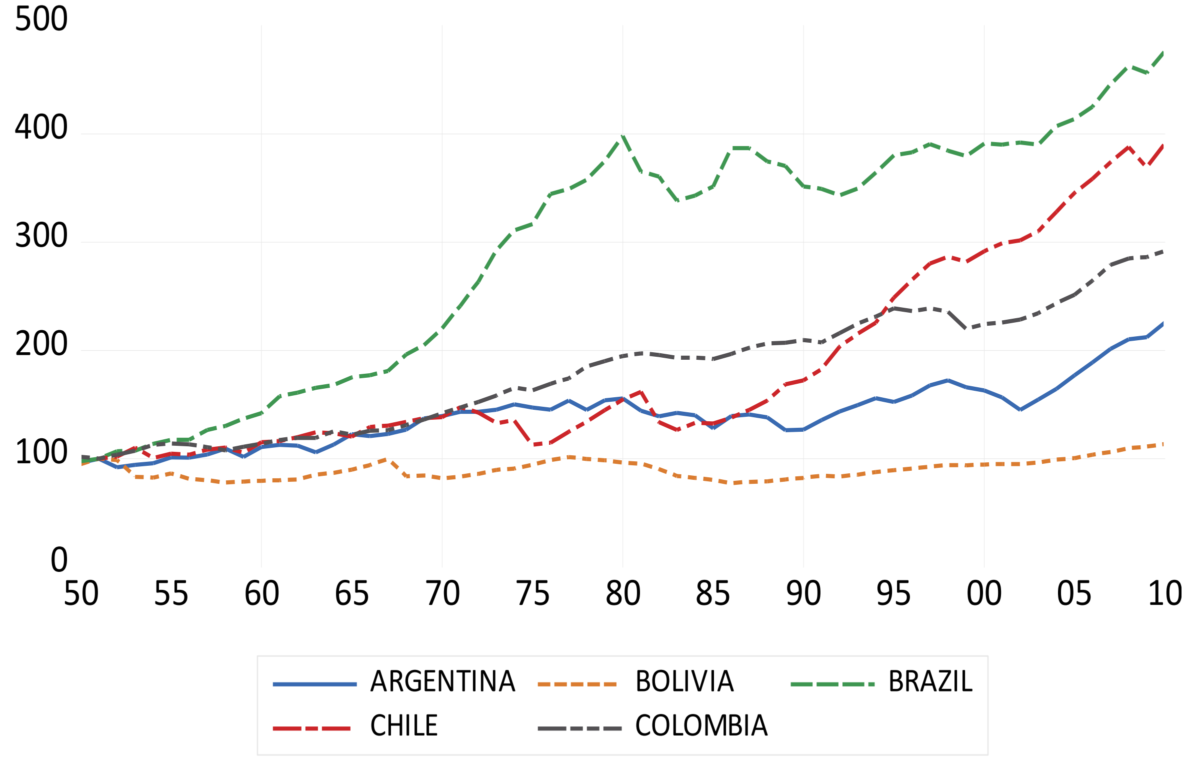
All observations until T are known, and with estimates of g and θ, the forecast becomes

$$y_{T+1|T} = g\left(y_T, y_{T-1}, \dots, y_{T-(p-1)}; \theta\right)$$

Typical features of economic time series

1. Trend (usually upward sloping)
2. Seasonality (different means and variances over the months/quarters within a year)
3. Aberrant observations (sudden and large changes in the mean due to extreme events)
4. Time-varying conditional variance (alternating periods with high and low volatility in financial markets)
5. Nonlinearity (different models in recessions and expansions)

1. Trend
real GDP per capita (1951 = 100)



What do we observe from this graph? Common growth? Structural breaks?

One way to describe a time series with a trend

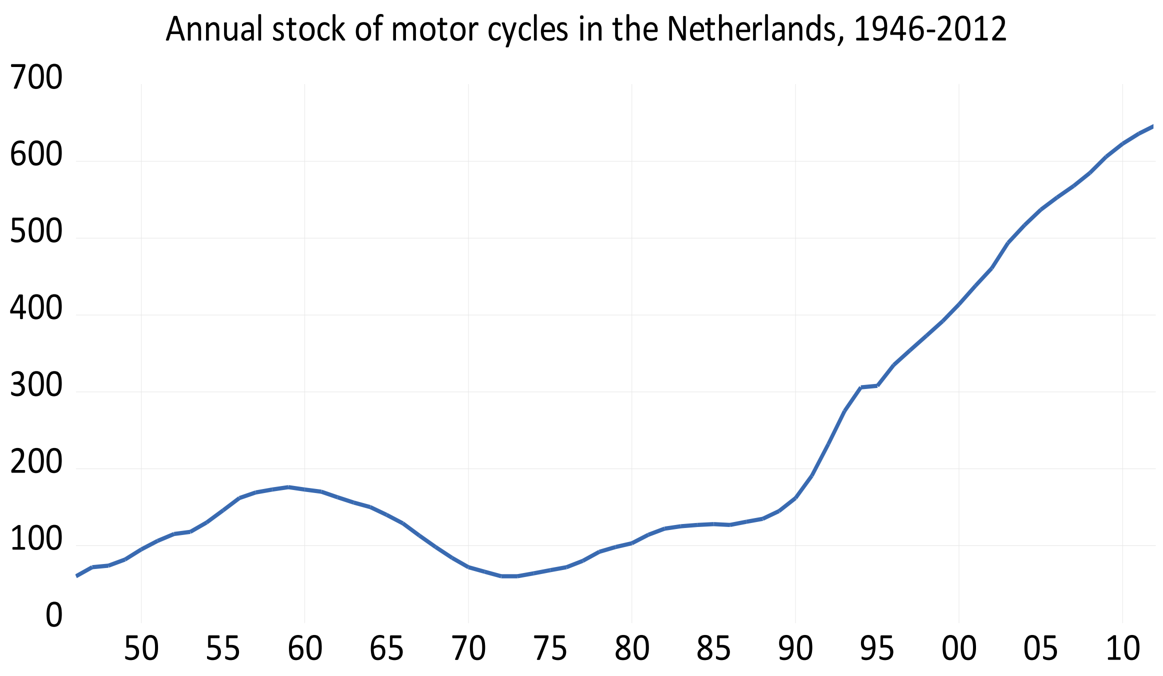
$$y_t = \alpha + \beta t + u_t$$

where $t = 1,2,3, \dots, T$. If we take y_t as the natural logs (ln) of the real GDP, Ordinary Least Squares (OLS) gives the estimates for β (with estimated standard errors)

Argentina	1.06 (0.07)
Brazil	2.48 (0.14)
Chile	2.28 (0.12)
Colombia	1.78 (0.04)
Mexico	1.92 (0.09)

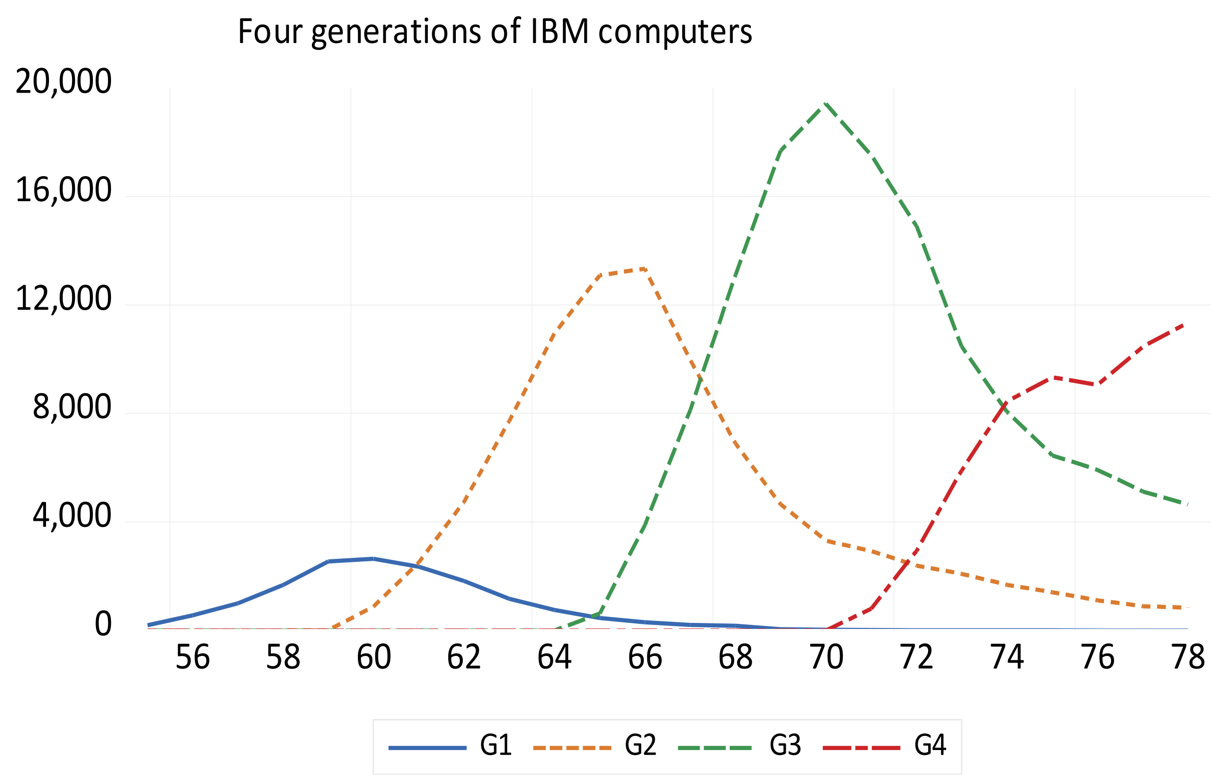
Common growth? Fastest growth?

The direction of a trend can change



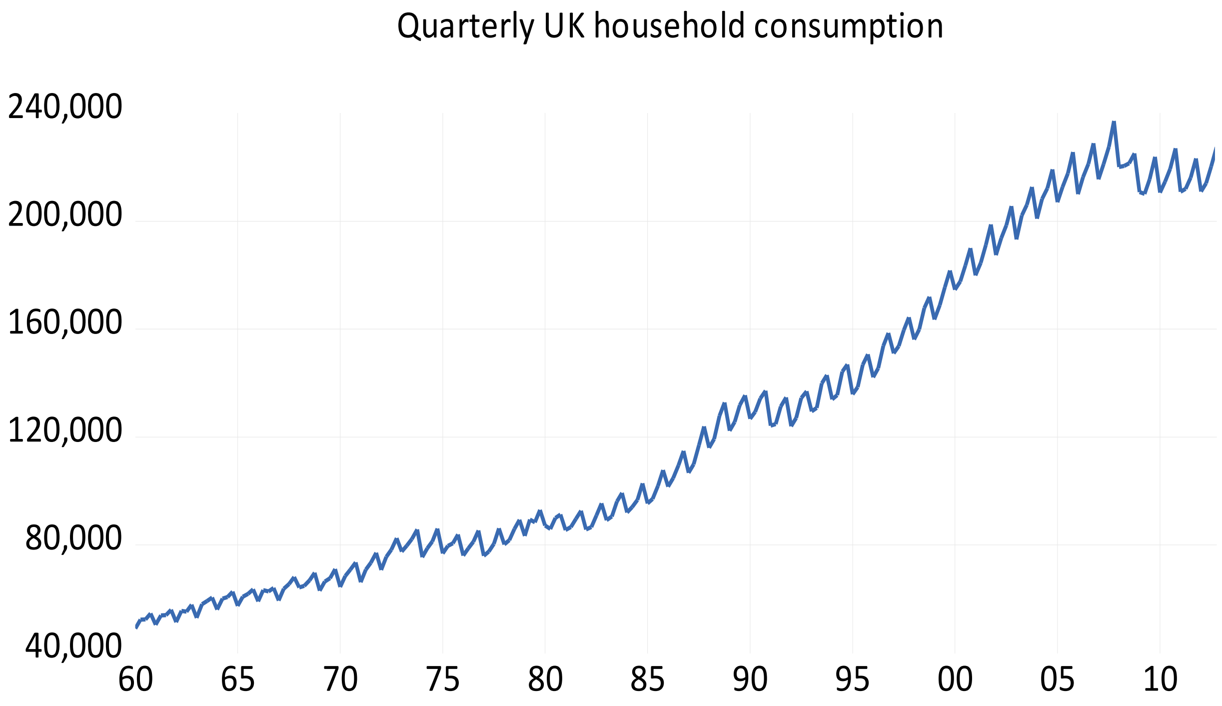
- What causes these changes?
- Are older data useful to predict the future?

Trends in life cycles



Where do sales eventually go to? What is important to predict?

2. Seasonality



Is seasonality constant (always peaks and troughs in same quarters)?

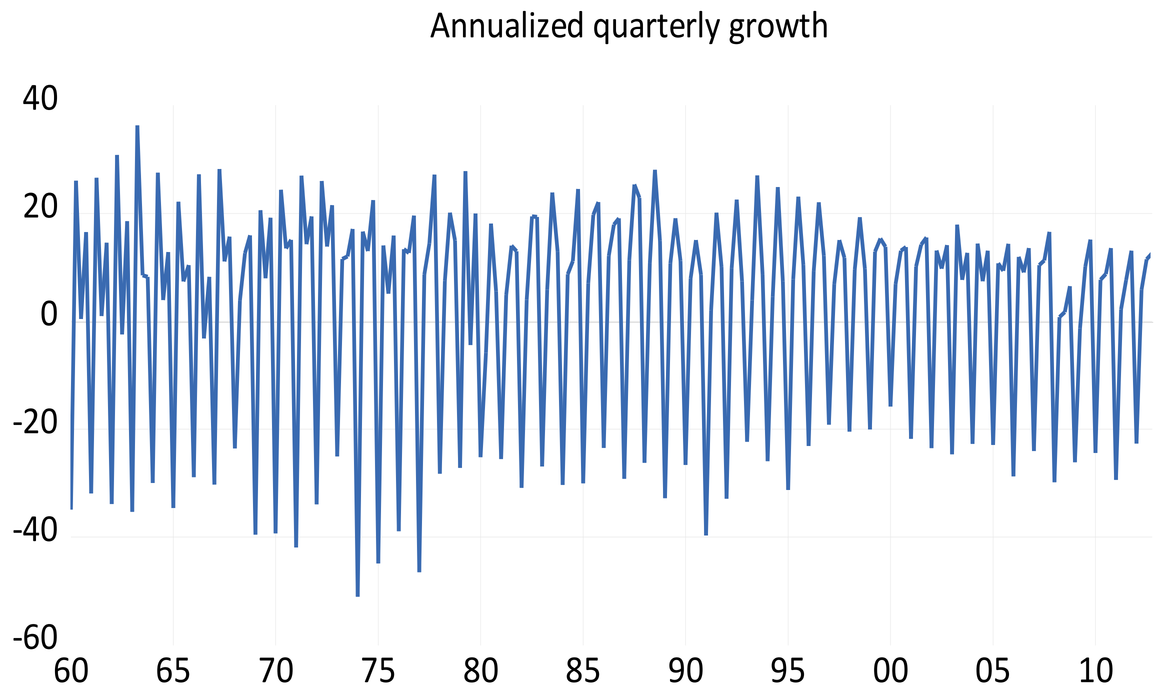
Growth rates for seasonal data

Change data into quarterly growth rates

$$100 * \ln\left(\frac{y_t}{y_{t-1}}\right)$$

Or into annualized quarterly growth rates

$$4 * 100 * \ln\left(\frac{y_t}{y_{t-1}}\right)$$



Is seasonality constant (always peaks and troughs in same quarters)?

A model for annualized quarterly growth rates

$$400 * \ln\left(\frac{y_t}{y_{t-1}}\right) = \alpha_1 D_{1,t} + \alpha_2 D_{2,t} + \alpha_3 D_{3,t} + \alpha_4 D_{4,t} + u_t$$

$$D_{1,t} = \begin{matrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ \vdots \end{matrix}, \quad D_{2,t} = \begin{matrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ \vdots \end{matrix}, \quad D_{3,t} = \begin{matrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ \vdots \end{matrix}, \text{ and } \quad D_{4,t} = \begin{matrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ \vdots \end{matrix}$$

Note

$$D_{1,t} + D_{2,t} + D_{3,t} + D_{4,t} = \begin{matrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{matrix}$$

So, alternatively:

$$400 * \ln\left(\frac{y_t}{y_{t-1}}\right) = \alpha_1 + (\alpha_2 - \alpha_1)D_{2,t} + (\alpha_3 - \alpha_1)D_{3,t} + (\alpha_4 - \alpha_1)D_{4,t} + u_t$$

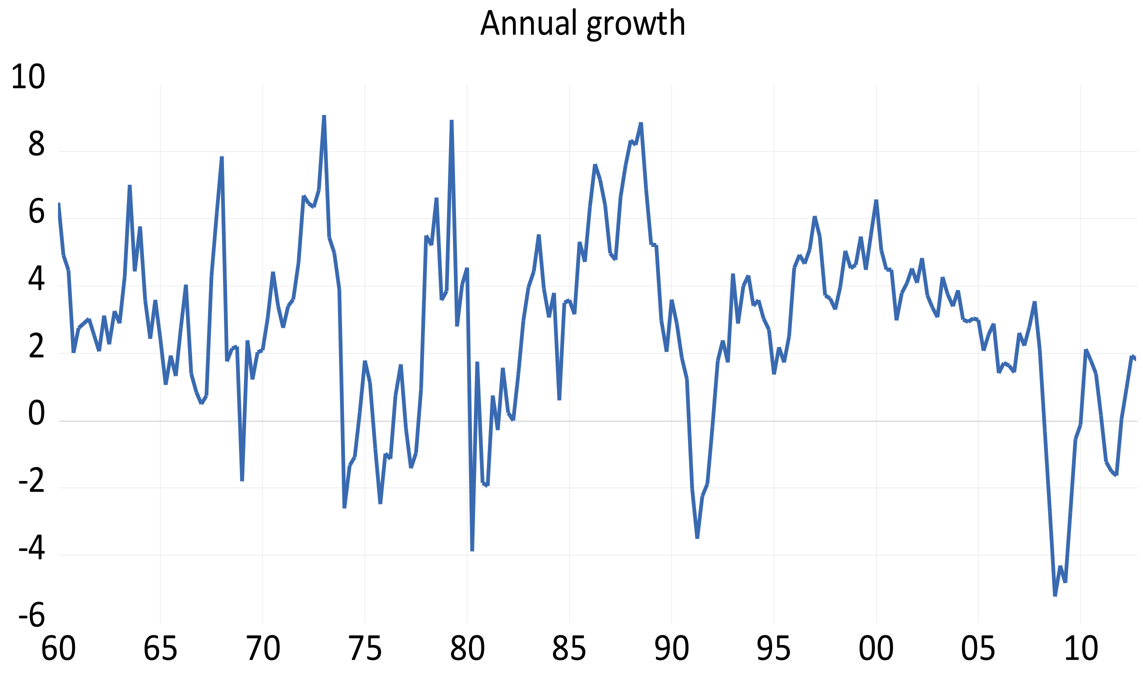
For UK consumption data:

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D1	-29.430	1.023	-28.758	0.000
D2	12.853	1.023	12.559	0.000
D3	12.913	1.023	12.618	0.000
D4	14.587	1.023	14.254	0.000
R-squared	0.864			

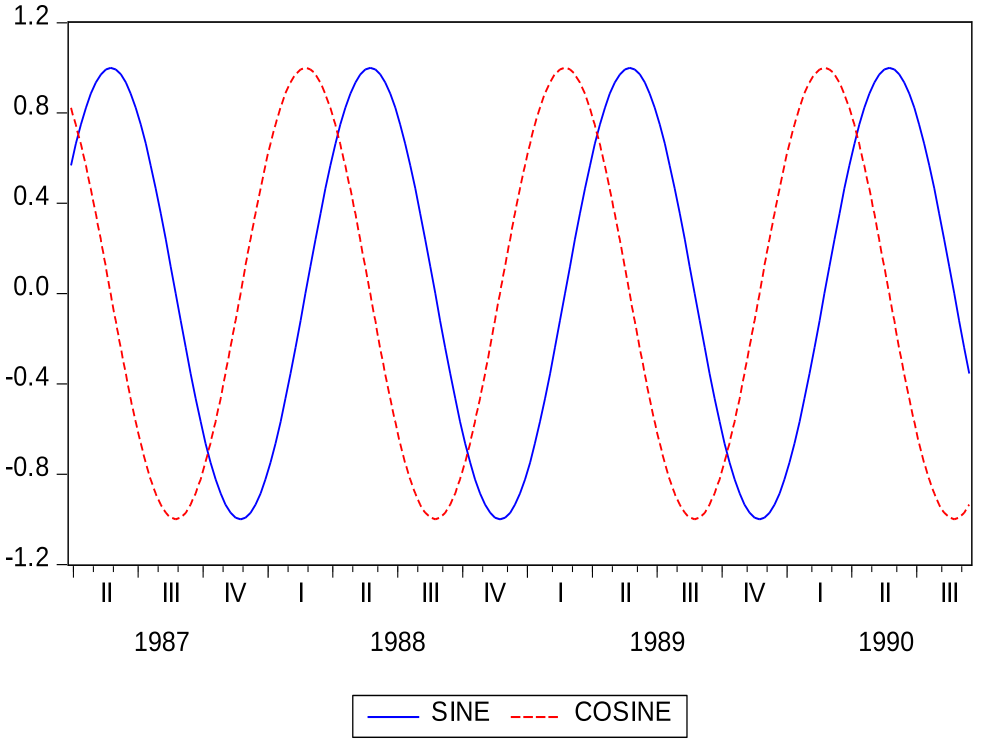
Another growth rate for seasonal data

Change data into annual growth rates

$$100 * \ln\left(\frac{y_t}{y_{t-4}}\right)$$

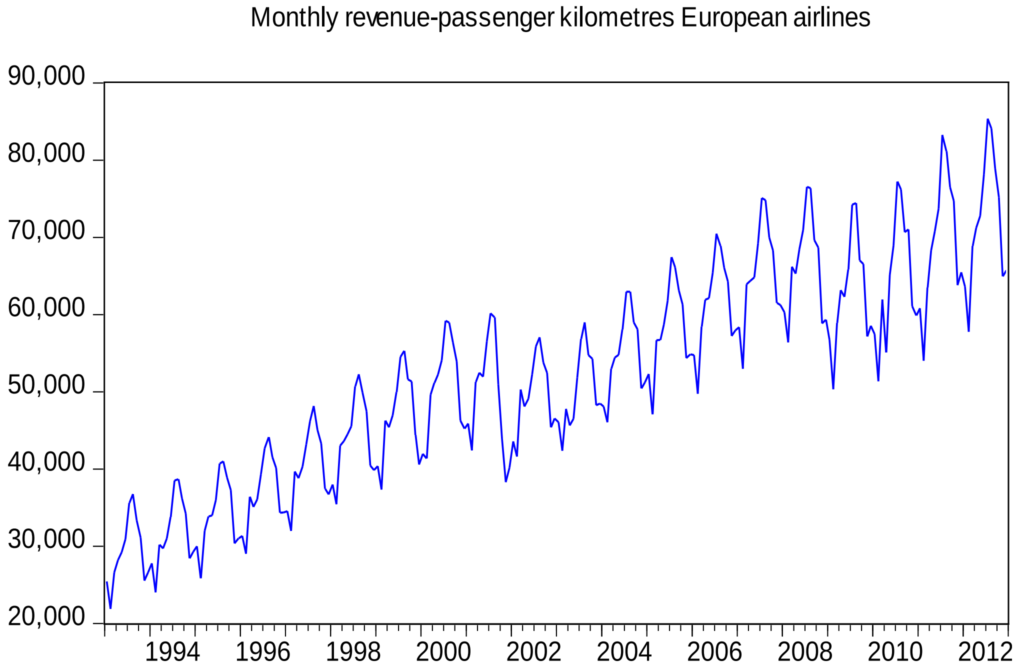


For higher frequency, like 52 weeks



$$100 * \ln\left(\frac{y_t}{y_{t-1}}\right) = \alpha_1 sin\left(\frac{2\pi t}{52}\right) + \alpha_2 cos\left(\frac{2\pi t}{52}\right) + u_t$$

3. Aberrant observations



What happened in 2001? And in 2008/2009?

Can impact be quantified?

In regression model with dummy

Change

$$y_t = \alpha + \beta t + u_t$$

into

$$y_t = \alpha + \beta t + \alpha^* D[t \geq 2011\text{M}09] + \beta^* D[t \geq 2011\text{M}09]t + u_t$$

with $D[t \geq 2011M09] = \cdots, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, \dots.$

In case of a single aberrant observation

Change

$$y_t = \alpha + u_t$$

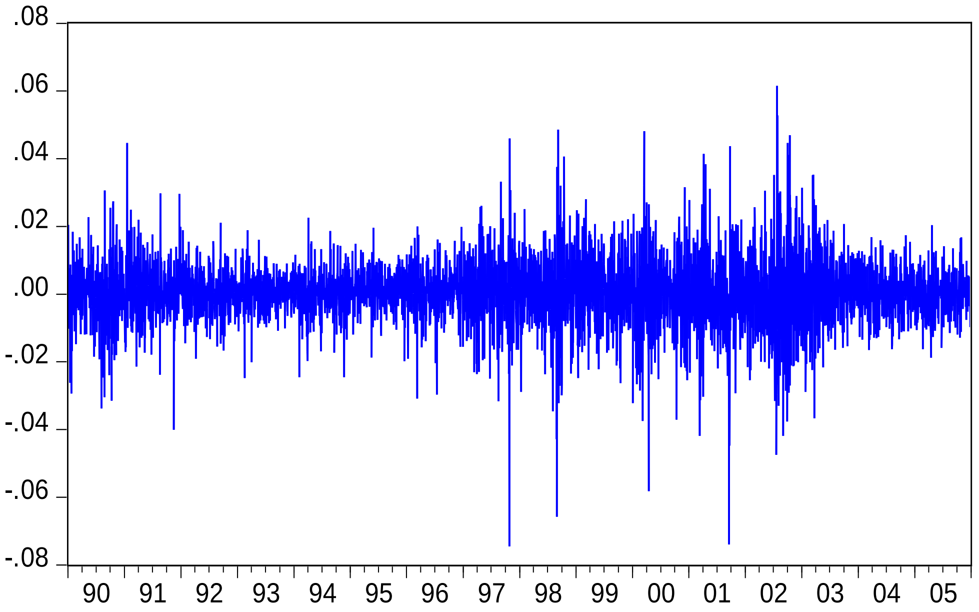
into

$$y_t = \alpha + \alpha^* D[t = \tau] + u_t$$

with $D[t = \tau] = \cdots, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, \dots.$

4: Conditional heteroskedasticity

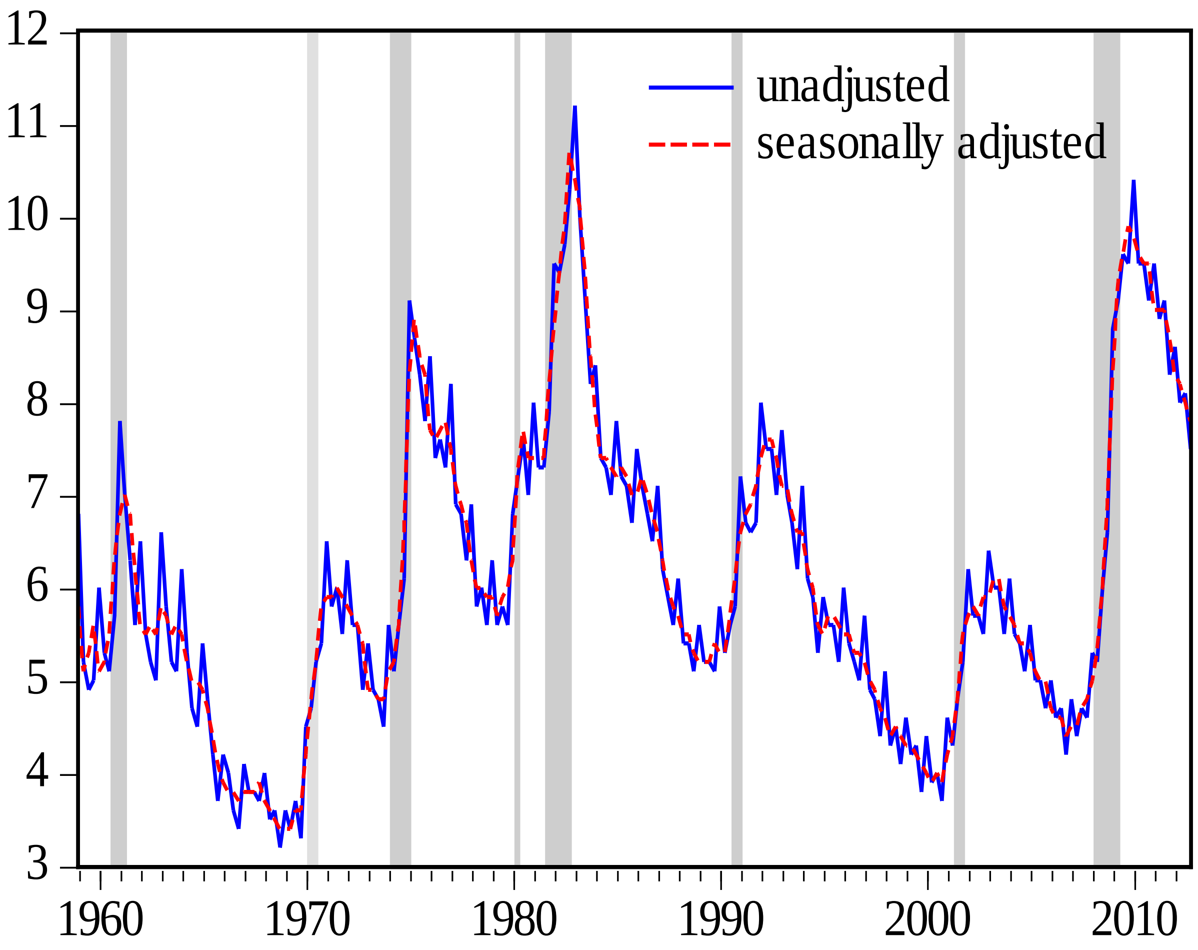
Daily returns on the Dow Jones index (1/1/1990-12/31/2005)



Normally distributed data? Correlation between extreme observations?

5. Nonlinearity

Unemployment rate US
(1959Q1-2012Q4)



Again, dummies

Change $$y_t - y_{t-1} = \mu + u_t$$

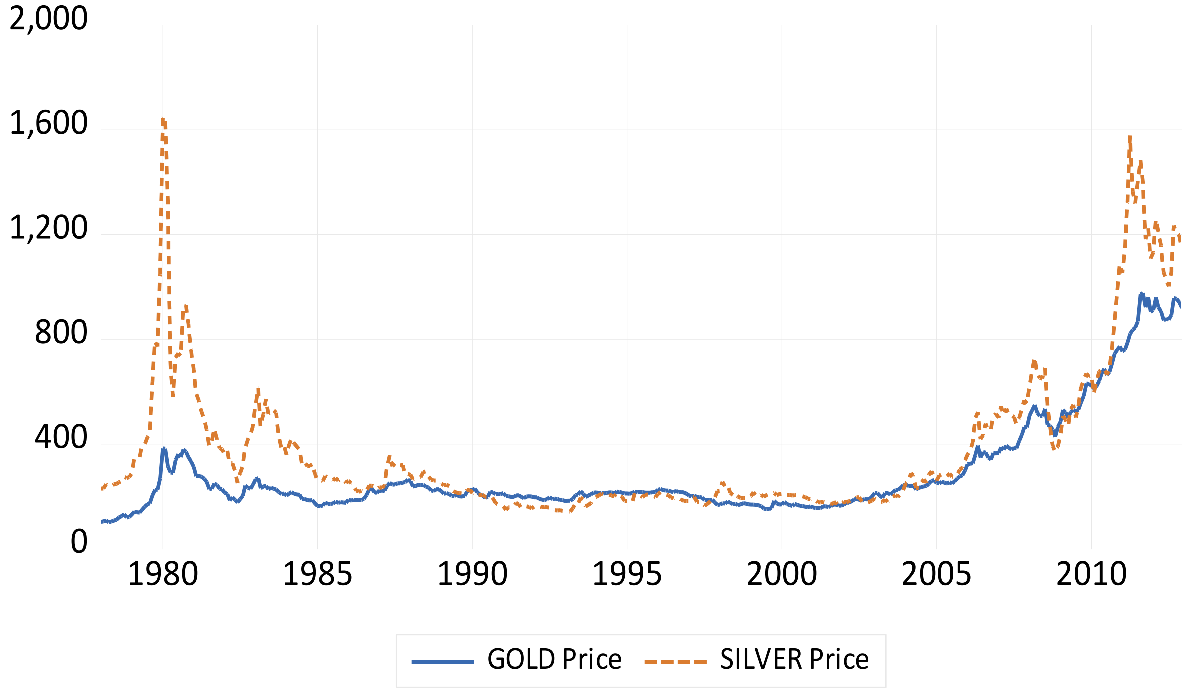
Into $$y_t - y_{t-1} = \mu_1 D[recession] + \mu_2 D(expansion) + u_t$$

with $D[recession] = \cdots, 0, 0, 0, 1, 1, 0, 0, 0, 0, 1, 1, 1, 1, 0, 0, 0, 0, \ldots$

For this sample:

Variable	Coefficient	Std. Error	t-Statistic	Prob.
Recession	0.581	0.044	13.179	0.000
Expansion	-0.093	0.018	-5.082	0.000

Extra: multivariate time series



Think of: $$y_t = \alpha + \beta x_t + u_t$$

Or perhaps better: $$y_t = \alpha + \beta x_{t-1} + u_t$$

What is time series analysis?

Start from the general form

$$y_t = g\left(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta\right) + \varepsilon_t$$

Find $g(.)$

Estimate θ

Check if ε_t cannot be predicted

If cannot be predicted, stop

If can be predicted, modify $g(.)$